MIL-D-80262A
20 June 1986
SUPERSEDING
MIL-D-80262
30 January 1984

MILITARY SPECIFICATION

DRILLING MACHINE, RADIAL, FLOOR MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor mounted, self-contained electric motor driven radial drilling machines suitable for general purpose drilling, tapping, boring, facing and reaming operations on large size or heavy workpieces with equipment and accessories specified herein.

1.2 Classification. The machines covered by this specification are of the plain radial arm type and are of the following sizes. The size furnished shall be as specified (see 6.2.1):

Size A4 - 9-inch column diameter by 3-feet arm length
Size A6 - 9-inch column diameter by 4-feet arm length
Size B1 - 11-inch column diameter by 3-feet arm length
Size B2 - 11-inch column diameter by 4-feet arm length
Size B3 - 11-inch column diameter by 5-feet arm length
Size C1 - 13-inch column diameter by 4-feet arm length
Size C2 - 13-inch column diameter by 5-feet arm length
Size D1 - 15-inch column diameter by 4-feet arm length
Size D2 - 15-inch column diameter by 5-feet arm length
Size D3 - 15-inch column diameter by 6-feet arm length
Size E1 - 17-inch column diameter by 5-feet arm length
Size E2 - 17-inch column diameter by 6-feet arm length
Size E3 - 17-inch column diameter by 7-feet arm length
Size F1 - 19-inch column diameter by 6-feet arm length
Size F2 - 19-inch column diameter by 7-feet arm length
Size F3 - 19-inch column diameter by 8-feet arm length

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3413
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

Size G1 - 22-inch column diameter by 7-feet arm length
Size G2 - 22-inch column diameter by 8-feet arm length
Size G3 - 22-inch column diameter by 9-feet arm length
Size G4 - 22-inch column diameter by 10-feet arm length
Size H1 - 26-inch column diameter by 8-feet arm length
Size H2 - 26-inch column diameter by 9-feet arm length
Size H3 - 26-inch column diameter by 10-feet arm length
Size H4 - 26-inch column diameter by 11-feet arm length
Size H5 - 26-inch column diameter by 12-feet arm length

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 79 - Electrical Standards for Industrial Machinery.

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B5.10 - Machine Tapers.

ANSI B11.8 - Drilling, Milling and Boring Machine Construction Care and Use of Safety Requirements.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

* (Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The drilling machine shall be new and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The drilling machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

* 3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the drilling machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2 Reclaimed materials. The drilling machine may contain reclaimed materials provided such materials will not jeopardize the drilling machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the drilling machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

3.2.3 Energy efficiency. The drilling machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the drilling machine that present safety hazards. The safety devices shall not interfere with the operation of the drilling machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All drilling machine parts, components, mechanisms, and assemblies furnished on the drilling machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. In addition, the machine shall comply with all the requirements of ANSI B11.8 that are designated therein as the responsibility of the machine manufacturers. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The drilling machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the drilling machine.

3.2.8 Environmental protection. The drilling machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the drilling machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation (see 6.4).

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The drilling machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the drilling machine to its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No processes such as welding, peening, plugging, or

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the drilling machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.065 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

3.3.8 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components.

3.4.1 Base. The base shall be a one-piece casting. The top of the base shall be machined for mounting the column and to provide a work surface. The work surface shall have not less than three, 5/8 inch or larger T-slots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

T-slots shall be in accordance with ANSI B5.1. A coolant fluid trough shall extend around the periphery of the work surface and be connected to a coolant reservoir which shall be integral with the base. A screen to prevent chips from entering the reservoir shall be furnished.

3.4.2 Plain type table. When specified (see 6.2.1), the machine shall have a plain type table configured and machined for attachment to the work surface of the base. The plain table shall be of box construction with finished top and side working surfaces. The top surface shall have three or more T-slots and the side surface two or more T-slots. T-slots shall be in accordance with ANSI B5.1.

3.4.2.1 Universal tilting type table. When specified (see 6.2.1), the machine shall have a universal tilting type table configured and machined for attachment to the work surface of the base. The table shall consist of a base on which shall be mounted a tilting work table with two machine finished work surfaces at 90° to each other. The top surface of the table shall be not less than 20 inches by 20 inches. The side work surface of the table shall be not less than 10 inches by 20 inches. The table surfaces shall be machined and provided with T-slots, three or more on the top and two or more on the side. T-slots shall be in accordance with ANSI B5.1. The table shall be arranged so that either work surface can be set in a vertical position by means of a worm and segment gear and have means for securing the work surfaces in position. The segment shall be graduated in degrees to indicate the angle at which the table is set. The base of the table shall be finished and provided with slots or bolt holes.

* 3.4.3 Column. The column shall consist of two ferrous castings, tubular in construction, with an inner member or trunk and an outer member or sleeve. The trunk shall be reinforced by internal ribs and shall have a flange for mounting on the base. The outer sleeve shall rotate 360° around the trunk on ball or roller bearings. The column shall be arranged with a clamping mechanism for securing the trunk to the sleeve binding the outer-sleeve to the trunk with a contact area of 360°. The finish on the surface giving support to the arm shall be not rougher than 16 rms.

* 3.4.4 Arm. The arm shall be a reinforced ferrous casting, machined and fitted to the column. The length of the arm bearing on the column shall be not less than twice the diameter of the column. Means shall be provided for lubricating the column sleeve when the arm is raised and lowered. Wipers shall be provided for cleaning the column sleeve. Ways shall be provided for movement of the head. The ways which provide the main support for the head shall have a surface hardness of not less than Rockwell C-50. The ways may be of the replaceable strip type. The arm shall have a rack gear for traversing the head along the arm. The arm shall rotate through 360° by means of the column sleeve rotation. Vertical arm movement shall be accomplished by an elevating screw and nut assembly driven either separately or by the main drive motor. The nut assembly shall be mounted on ball or roller bearings and shall be equipped with a safety device that supports the arm in case of thread stripping. Automatic limit stops shall be provided to

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

arrest the travel of the arm at both ends of the vertical range. The elevating screw shall be supported at both ends with support brackets and bearings and affixed to the column at both top and bottom. A clamping device shall be furnished for securing the arm to the column. When specified (see 6.2.1), machines with column diameters larger than 13 inches shall have either electric, hydraulic, or electrohydraulic clamping devices for securing the arm to the column and the column trunk to the column sleeve.

* 3.4.5 Head. The head shall be fully enclosed, machined and fitted to the arm. A clamp shall be provided for locking the head to the arm. The head shall be traversed along the arm by means of a power or manually operated rack and pinion gear mechanism. When specified (see 6.2.1), the heads of machines with column diameters larger than 10 inches shall be traversed by means of a power operated mechanism. When power traverse is provided, the head shall incorporate an interlock or slip clutch to prevent damage from engagement of the rapid traverse while the head is in a locked position. The spindle and all rotating member sliding surface sleeves, bearings and component shall be lubricated by a constant supply of oil during use.

3.4.6 Spindle. The spindle shall be of heat-treated alloy steel with a surface hardness of not less than Rockwell C-40. The spindle shall be mounted in a sleeve which shall support or drive the spindle. The spindle shall be supported by ball or roller bearings. The spindle counterbalance mechanism shall be fully enclosed in the head. The spindle taper shall be of the size specified in table I, and shall comply with the requirements of ANSI B5.10.

3.4.7 Speeds and feeds The spindle speed and feed change controls shall be provided on the head. Speed changes shall be controlled by not more than three conveniently located levers and accomplished through sliding gears and positive tooth clutches. All speed and feed gears shall be of heat-treated alloy steel. Sliding gears shall slide on multiple hardened and ground involute splined shafts having not less than three splines with integral keys. A depth gage shall be provided which shall automatically disengage the power feed at any predetermined depth indicated on the graduated dial. A device shall be included to automatically disengage the feed when the limit of spindle travel is reached. The feed engagement shall be operated by a positive tool or friction type clutch with means for adjustment.

3.4.8 Coolant system. The coolant system shall include a sump or reservoir, a power-driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.9 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A fliter system shall be provided to ensure delivery of

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

clean fluid. The hydraulic reservoir shall have means for determining fluid level.

* 3.4.10 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.10.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for dripproof enclosure.

3.4.10.2 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.4.10.3 Solid-state components. Solid-state design shall be used throughout for electronic components.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and characteristics of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. The machine shall be capable of performing drilling, tapping, boring, and reaming operations for accomplishing all functional characteristics for which it is designed along with those required by table I. The machine shall also be capable of performing full horsepower drilling without any evidence of chatter, stalling, or permanent distortion.

3.7 Alignment and accuracy. The machine shall meet the alignment and accuracy tolerances required by table II.

3.8 Standard equipment. Unless otherwise specified, all standard equipment normally furnished by the manufacturer shall be furnished. Additional equipment shall be furnished as specified and fully described (see 6.2.1).

3.9 Optional equipment. Optional equipment specified in the following subparagraphs shall be furnished with each machine, as specified (see 6.2.1).

a. A tapping attachment, unless a tapping feature is built into the machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

b. One worklight, to illuminate the work area, complete with protective shield.

3.10 Additional requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified (see 6.2.1).

3.11 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.12 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine and contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.13 Hour meter. Unless otherwise specified (see 6.2.1), the drilling machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.14 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.15 Workmanship. Workmanship of the drilling machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing amoung manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or all tests shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. Failure of the item to pass the examination, the tests, or the inspection shall be cause for rejection.

4.6 Examination. The drilling machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The drilling machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Alignment tolerance test. Each drilling machine shall be tested to determine that it complies with the alignment tolerance requirements of 3.7.

4.7.3 Performance tests. Each machine shall drill 15 through holes, of the maximum rated size for the machine in 1-inch thick soft cast iron. For this test drill size shall be 2 inches for 9-inch column machines, 2-1/4 inches for 11-inch column machines and 4 inches for all other column diameter machines covered by the specification. Drilling time shall be not more than 20 minutes, total elapsed time. There shall be no evidence of overloading, overheating, excess vibration, or deformation of part of the machine during or after drilling operation.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The radial drilling machines covered by this specification are intended for drilling, reaming, spot facing, tapping, boring and counterboring operations on work too large or too heavy to be placed on an upright drilling machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Sizes required (see 1.2).

c. First article, if required (see 3.1, 4.4, and 6.3).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. Specify additional safety and health requirements (see 3.2.5).

f. Painting, if different (see 3.3.5).

g. If electromagnetic interference control is required, specify the equipment and subsytem class and the emission and susceptibility required (see 3.3.9).

h. Plain table, if required (see 3.4.2).

i. Universal tilting table, if required (see 3.4.2.1).

j. Powered clamping device, if required (see 3.4.4).

k. Type of traversing mechanism, if different (see 3.4.5).

l. Electrical system and input power, if different (see 3.4.10).

m. Type motor enclosure, if different (see 3.4.10.1).

n. Size and characteristics, if different (see 3.5).

o. Standard equipment, if different (see 3.8).

p. Specify and fully describe optional equipment, if required (see 3.9).

q. Performance requirements, if different (see 3.10).

r. Lubrication chart or plate, if different (see 3.11).

s. Nameplate, if different (see 3.12.2).

t. Hour meter, if different (see 3.13).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

u. Quality conformance inspection, if different (see 4.4).

v. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the drilling machine itself. It does not require compliance with those OSHA requirements that concern "the drilling machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual drilling machines in the environment), the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirement that will integrate the new machine into its future operating environment. The drilling machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, since the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1.e.

6.5 Subject terms (key word) listing:

Drilling machine radial arm
Boring
Drilling
Facing
Floor mounted
Heavy duty
Reaming
Tapping

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities.

Characteristics	Size A4	Size A6	Size B1	Size B2	Size B3	Size C1	Size C 2	Size D1	Size D2
Column diameter, inches	9	9	11	11	11	13	13	15	15
Arm length, feet	3	4	3	4	5	4	5	4	5
Spindle to base, maximum, inches	53	53	55	55	55	60	60	66	66
Spindle to base, minimum, not more than, inches	14	14	14	14	14	11	11	11	11
Arm bearing length on column, inches	18	18	22	22	22	26	26	30	30
Arm traverse on column, inches	26	26	25	25	25	29	29	35	35
Spindle traverse in head, inches	10	10	12	12	12	18	18	18	18
Head traverse on arm, inches	25	37	25	37	49	32	44	32	44
Spindle center to column, minimum, not more than, inches	13	13	11	11	11	16	16	16	16
Base height, inches	6	6	8	8	8	8	8	8	8
Base working length, inches	41	53	41	53	65	53	65	53	65
Base working width, inches	30	30	36	36	36	36	36	40	40
Spindle speeds, number of	9	9	12	12	12	32	32	32	32
Spindle speed range, RPM	100 to 1500	100 to 1500	80 to 1500	80 to 1500	80 to 1500	15 to 1200	15 to 200	15 to 1200	15 to 1200
Spindle feeds, number of	4	9	6	6	6	16	16	16	16
Spindle feed range, inch per revolution	.004 to .020	.004 to .020	.003 to .020	.003 to .020	.003 to .020	.004 to .125	.004 to .125	.004 to .125	.004 to .125
Spindle diameter at point of drive, inches, minimum	1-3/8	1-3/8	1-1/2	1-1/2	1-1/2	2-1/2	2-1/2	2-3/16	2-3/16
Motor horsepower	5	5	5	5	5	10	10	15	15
Taper in spindle, Morse No:	4	4	4	4	4	5	5	5	5
Tape leads, threads per inch:									
1	-	-	-	-	-	8	8	8	8
2	-	-	-	-	-	11-1/2	11-1/2	11-1/2	11-1/2
3	-	-	-	-	-	14	14	14	14
4	-	-	-	-	-	18	18	18	18

NOTE: Drilling machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size as shown in above table. When the largest size and capacity machine shown in the above table or ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities (continued).

Characteristics	Size DJ	Size E1	Size E2	Size E3	Size F1	Size F2	Size F3	Size G1
Column diameter, inches	15	17	17	17	19	19	19	22
Arm length, feet	6	5	6	7	6	7	8	7
Spindle to base, maximum, inches	66	72	72	72	78	78	78	96
Spindle to base, minimum, not more than, inches	11	15	15	15	15	15	16	22
Arm bearing length on column, inches	30	34	34	34	38	38	38	44
Arm traverse on column, inches	35	37	37	37	43	43	43	50
Spindle traverse in head, inches	18	18	18	18	18	18	18	24
Head traverse on arm, inches	56	47	58	70	57	79	79	67
Spindle center to column, minimum, not more than, inches	16	15	15	15	15	15	15 to 17	17
Base height, inches	8	10	10	10	10	10	10	12
Base working length, inches	77	65	77	89	77	89	101	90
Base working width, inches	40	50	50	50	50	50	50	57
Spindle speeds, number of	32	32	32	32	32	32	32	36
Spindle speed range, RPM	15 to 1200	15 to 1200	15 to 1200	15 to 1200	15 to 1200	15 to 1200	15 to 1200	8 to 800
Spindle feeds, number of	16	16	16	16	16	16	16	18
Spindle feed range, inch per revolution	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125
Spindle diameter at point of drive, inches	2-3/16	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2	3
Motor horsepower	15	15	15	15	20	20	20	20
Taper in spindle, Morse No.	5	5	5	5	6	6	6	6
Tape leads, threads per inch:								
1	8	8	8	8	8	8	8	8
2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2
3	14	14	14	14	14	14	14	14
4	18	18	18	18	18	18	18	18

NOTE: Drilling machines sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size as shown in above table. When the largest size and capacity machine shown in the above table or ordered, the size and capacitiy of the machine offered shall not exceed the stated requirements by more than 10 percent. Whne a range is shonw, the required performance is from the stated minimum or less to the stated maximum or greater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities (continued).

Characteristics	Size G2	Size G3	Size G4	Size H1	Size H2	Size H3	Size 4	Size H5
Column diameter, inches	22	22	22	26	26	26	26	26
Arm length, feet	8	9	10	8	9	10	11	12
Spindle to base, maximum, inches	96	96	96	100	100	100	100	100
Spindle to base, minimum, not more than, inches	22	22	22	22	22	22	22	22
Arm bearing length on column, inches	44	44	44	52	52	52	52	52
Arm traverse on column, inches	50	50	50	50	50	50	50	50
Spindle traverse in head, inches	24	24	24	24	24	24	24	24
Head traverse on arm, inches	79	91	103	80	92	104	116	128
Spindle center to column, minimum, not more than, inches	17	17	17	22	22	22	22	22
Base height, inches	12	12	12	12	12	12	12	12
Base working length, inches	102	114	126	102	114	126	138	150
Base working width, inches	57	57	57	57	57	57	57	57
Spindle speeds, number of	36	36	36	36	36	36	36	36
Spindle speed range, RPM	8 to 800	8 to 800	8 to 800	8 to 800	8 to 800	8 to 800	8 to 800	8 to 800
Spindle feeds, number of	18	18	18	18	18	18	18	18
Spindle feed range, inch per revolution	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125
Spindle diameter at point of drive, inches	3	3	3	3	3	3	3	3
Motor horsepower	20	20	20	20	20	20	20	20
Taper in spindle, Morse No:	6	6	6	6	6	6	6	6
Taps leads, threads per inch:								
1	8	8	8	8	8	8	8	8
2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2
3	14	14	14	14	14	14	14	14
4	18	18	18	18	18	18	18	18

NOTE: Drilling machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size as shown in above table. When the largest size and capacity machine shown in the above table or ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimjm or less to the stated maximum or greater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

6.6 Changes from previous issue. The margins of the specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements in this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

TABLE II. Alignment and accuracy tolerances.

Components	Maximum allowance deviation
Work surface, flat in longitudinal direction	0.0012 inch per foot
Work surface, flat in traverse direction	0.0012 inch per foot
traverse column	
1/ Column square with base in plane through center line of base	
For machines up to 4 feet 7 inches length radial arm	0 to 0.0024 inch per foot
For machines over 4 feet 7 inches length of radial arm	0 to 0.0036 inch per foot
Column square with base in plane perpendicular to plane through center line of baseplate	0.0012 inch per foot
Arm	
1/ Arm parallel with base (low at outer end)	0 to 0.0024 inch per foot
For machines up to 4 feet 7 inches length	0 to 0.0036 inch per foot
For machines over 4 feet 7 inches length of radial arm	0 to 0.0024 inch per foot
Arm ways, flat	
Drilling spindle	
Spindle runout (using 12-inch long mandrel 12 inches from spindle nose	0.002 inch
Spindle square with base in plane through center line of base (lower end inclined toward column)	0 to 0.0024 inch per foot
Spindle square with work surface of base in plane 90 degrees to above plane	0.0012 inch per foot
Travel of spindle sleeve square with work surface of base in plane through center line of base (inclined at lower end toward column)	0 to 0.004 inch per foot
Travel of spindle square with base in plane 90 degrees to above plane	0.002 inch per foot
2/ Drill spindle square with plain type table mounted on base in plane through center line of base (spindle inclined at lower end toward column)	0 to 0.0024 inch per foot
2/ Drill spindle square with plain type table mounted on base in plane perpendicular to above plane	0.0012 in per foot

NOTE: 1/ When testing, the arm shall be positioned half-way up the column and the head half-way along the arm.

2/ When a plain type table is furnished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activities:
Army - SM
Navy - YD, MC

Preparing activity:
DLA - IP

Project (3413-0138)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-80262A	Drilling Machine, Radial, Floor Mounted

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:52Z
Check the source to verify that this is the current version before use.